Pandita's Aether-DCTWIN

Smart Cooling, Capacity & Uptime Software for Data Center Facilities

The Problem Today

- Data centers mix **15 kW air-cooled racks** with **80 kW+ liquid-cooled AI racks**, so teams run cooling colder and faster than needed just to stay safe.
- That safety buffer wastes **10–20% on cooling power** and leaves paid-for power and cooling capacity sitting unused.
- Legacy BMS screens only show current temperatures—not whether a row will survive a pump trip, or if a new 500 kW customer can fit next month.

How It Works (100% Facility-Side)

STEP 1

Read Facility Meters

Connects to existing BMS, cooling & row PDU meters (or CSV exports). **Zero software on customer servers.**

→

STEP 2

Model the Floor Live

Learns your rows in 48 hours to predict heat spikes, equipment failures, and spare rack capacity.

→

STEP 3

Tune Safely

Starts 100% read-only (Shadow Mode). Live tuning auto-falls back to your local BMS if disconnected.

Four Everyday Outcomes

8% – 18% Less

Cooling & Peak Power Cost

Slows over-spinning fans and pumps, uses outdoor free cooling, and shaves expensive peak utility hours.

+10% – 20% More

Usable Rack Capacity

Finds capacity stranded on unbalanced 3-phase PDUs and validates new customer fit-outs in 60 seconds.

Zero Surprises

Uptime & Leak Protection

Shows live seconds-to-limit if any cooling unit trips (N-1), and keeps liquid pipes above dew point so they never sweat.

90% Less

NOC Alarm Fatigue

Groups 50+ rack alerts into 1 mechanical root cause, predicts filter clogging, and writes shift handover logs.

What's Included (16 Capabilities in 4 Simple Areas)

1. Uptime & Safe Control

- ✓ Read-only Shadow Mode & safe BMS fallback
- ✓ Live N-1 cooling trip countdown clock
- ✓ Liquid pipe dew-point & condensation guard
- ✓ Step-by-step maintenance checklists (MOPs)

2. Capacity & Colo Sales

- ✓ 60-second customer fit-out & summer check
- ✓ 3-phase PDU breaker balance & whip advisor
- ✓ Tenant SLA proof & monthly ESG/water reports
- ✓ 3D floor view & offline "What-If" sandbox

3. Lower Energy Bills

- ✓ Smooth CRAC fan & CDU coolant tuning
- ✓ Outdoor wet-bulb free-cooling advisor
- ✓ Peak utility tariff & UPS battery shaving
- ✓ Plain-dollar ROI & kWh savings dashboard

4. Simpler Daily Ops

- ✓ 48-hour automatic hall thermal calibration
- ✓ Filter & strainer replacement timing
- ✓ 50-to-1 root-cause alarm grouping & tickets
- ✓ One-click shift handover & 24-hour replay

Example Value on One 5 MW Data Hall

~$700K / yr
Direct cooling electricity, peak-demand & maintenance savings *(payback under 5 months)*.

+650 kW
Stranded capacity reclaimed —worth **$5.5M** in avoided build cost or **+$1.1M/yr** in rack leases.

0 Agents
Runs on-prem in your facility network using standard BACnet, Modbus & SNMP.

Zero-Risk 30-Day Pilot

WEEK 1 • CONNECT
Upload a 30-day BMS/PDU CSV or connect read-only meters. Zero changes on the floor.

WEEKS 2–3 • PROVE
Review live Shadow savings, N-1 safety countdowns, and stranded 3-phase PDU kW.

WEEK 4 • VERIFY
Test supervised cooling recommendations on one zone and confirm the meter drop.